Machine Learning Laboratory
Experiment No. 08
Experiment Name: Linear Support Vector Machine (SVM)

Aim

Implement a Linear Support Vector Machine (SVM) to classify the Iris dataset. Visualize the decision boundary and discuss how the margin is determined.

Objectives

- Load and preprocess the Iris dataset.
- Implement a Linear SVM for binary classification (e.g., classify Setosa vs. NonSetosa).
- Visualize the decision boundary and margin.
- Discuss the concept of the margin and how it influences classification.

Theory

Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification and regression problems. For classification, SVM finds an optimal decision boundary, called a hyperplane, that separates the classes.

For a two-dimensional feature space, the linear decision boundary can be represented as:

$$w_1 x_1 + w_2 x_2 + b = 0$$

where,

- w_1 and w_2 are the weights.
- x_1 and x_2 are the input features.
- b is the bias term.

The decision function is given by:

$$f(X) = w^T X + b$$

The predicted class is determined as:

$$\hat{y} = \begin{cases} +1, & w^T X + b > 0 \\ -1, & w^T X + b < 0 \end{cases}$$

Decision Hyperplane

The decision hyperplane is the boundary that separates the different classes. For a Linear SVM, the hyperplane is:

$$w^T X + b = 0$$

For two features:

$$w_1 x_1 + w_2 x_2 + b = 0$$

The SVM selects the hyperplane that provides the maximum separation between the classes.

Margin

The margin is the distance between the decision boundary and the closest data points from each class.

The two margin boundaries are given by:

$$w^T X + b = +1$$

and

$$w^T X + b = -1$$

The distance between these two boundaries is:

$$\text{Margin} = \frac{2}{\|w\|}$$

where,

$$\|w\| = \sqrt{w_1^2 + w_2^2}$$

Therefore, the SVM tries to maximize the margin. This is equivalent to minimizing:

$$\frac{1}{2}\|w\|^2$$

A larger margin generally provides better generalization and makes the classifier less sensitive to small variations in the training data.

Support Vectors

Support vectors are the training data points that lie closest to the decision boundary. These points determine the position and orientation of the optimal hyperplane.

The support vectors lie on or closest to the margin boundaries:

$$w^T X + b = +1$$

and

$$w^T X + b = -1$$

Thus, the support vectors play an important role in determining the maximum-margin decision boundary.

Iris Dataset

The Iris dataset contains measurements of iris flowers belonging to three different species:

- Setosa
- Versicolor
- Virginica

The dataset contains four features:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

For this experiment, the first 100 samples corresponding to Setosa and Versicolor classes are selected, and the first two features are used for classification and visualization.

The selected features are:

- Sepal Length
- Sepal Width

Feature Scaling

Feature scaling is a preprocessing technique used to bring the input features into a common range. It prevents features with larger numerical values from dominating the learning process.

Standardization is performed using:

$$X_{\text{scaled}} = \frac{X - \mu}{\sigma}$$

where,

- μ represents the mean of the feature.
- σ represents the standard deviation of the feature.

After standardization, each feature has approximately:

$$\text{Mean} = 0, \qquad \text{Standard Deviation} = 1$$

Linear SVM

A Linear SVM uses a linear kernel to find a straight decision boundary between the classes.

In Scikit-learn, a Linear SVM can be implemented using:

```
SVC(kernel="linear")
```

The Linear SVM learns the weights w and bias b from the training data.

Performance Metrics

Accuracy

Accuracy represents the percentage of correctly classified samples.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

For binary classification:

- TP = True Positive
- TN = True Negative
- FP = False Positive
- FN = False Negative

Algorithm

Step 1: Start.

Step 2: Import the required libraries such as NumPy, Pandas, Matplotlib, and Scikit-learn modules.

Step 3: Load the Iris dataset.

Step 4: Select the first 100 samples corresponding to Setosa and Versicolor classes and choose the first two features.

Step 5: Store the selected features in X and class labels in y.

Step 6: Split the dataset into training and testing sets.

Step 7: Create a Linear Support Vector Machine (SVM) classifier using a linear kernel.

Step 8: Train the SVM model using the training data.

Step 9: Use the trained model to predict the class labels of the test data.

Step 10: Evaluate the model performance by calculating the accuracy and classification report.

Step 11: Determine the minimum and maximum feature values and create a mesh grid for plotting.

Step 12: Predict the class labels for all points in the mesh grid to obtain the decision regions.

Step 13: Plot the decision boundary and classified regions using the predicted mesh grid values.

Step 14: Highlight the support vectors and display the dataset points along with the decision boundary and margin.

Step 15: Stop.

Expected Outcome

- The Iris dataset is successfully loaded and preprocessed.
- A Linear SVM successfully classifies Setosa and Versicolor classes.
- The decision boundary and margin are successfully visualized.

- The support vectors are identified and highlighted in the plot.
- The SVM determines the optimal separating hyperplane by maximizing the margin between the classes.
- The model performance is evaluated using classification accuracy and a classification report.

Result

A Linear Support Vector Machine (SVM) was successfully implemented on the Iris dataset using the first two features of the first 100 samples. The model successfully classified Setosa and Versicolor classes. The decision boundary, margin, and support vectors were visualized. The SVM determines the optimal decision boundary by maximizing the margin between the classes.